NMP Project 01: Vector and Matrix Operations

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1 Implementation Details

The dynamic constructor initializes matrix bounds and memory layout. Here we show both the semantic title and the real source location (lines extracted from the file relative to the project root):

`Matrix Dynamic Constructor` `include/Matrix.hpp:19-36`

```
Matrix() : n(0), m(0), lda(0), _data() {}

template <typename Function>
Matrix(size_t n, size_t m, Function f)
    : Matrix(n, m, 0.0)
{
    for (size_t i = 0; i < n; ++i)
    {
        for (size_t j = 0; j < m; ++j)
        {
            _data[row_offset(i, j, lda)] = f(i, j);
        }
    }
}
```

2 Matrix Rows Accessor

The rows accessor retrieves row dimensions efficiently. Here we demonstrate referring to the snippet by its **real location only** with line numbers disabled:

`src/Matrix.cpp:19-22`

```
size_t Matrix::rows() const
{
    return n;
}
```

3 Scoped Environment Example

You can also scope a specific root to an environment block:

The BLAS-accelerated multiplication kernel inside a scoped environment (bracket reference format):

`Matrix::multiply_blas [src/Matrix.cpp:138-163]`

```
Matrix Matrix::multiply_blas(const Matrix &other) const
```

```
{
    if (m != other.n)
    {
        throw std::invalid_argument("Incompatible dimensions for multiplication");
    }

    Matrix result(n, other.m, 0.0);
    cblas_dgemm(
        CblasRowMajor,
        CblasNoTrans,
        CblasNoTrans,
        static_cast<int>(n),
        static_cast<int>(other.m),
        static_cast<int>(m),
        1.0,
        _data.data(),
        static_cast<int>(lda),
        other._data.data(),
        static_cast<int>(other.lda),
        0.0,
        result._data.data(),
        static_cast<int>(result.lda)
    );
    return result;
}
```

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4 Cross-Referencing Snippets in Document Prose

Snippets can be referenced in body text by their semantic name, real location, or line range:

- Constructor: `Matrix Dynamic Constructor` is defined in `include/Matrix.hpp:19-36` (lines 19-36 of `include/Matrix.hpp`).
- Accessor: `Matrix::rows` is located at `src/Matrix.cpp:19-22`.
- BLAS Kernel: `Matrix::multiply_blas` is located at `src/Matrix.cpp:138-163`.